

Two challenges, one laser

Entangling two atoms at the quantum speed limit, then making five atoms solve an NP-hard puzzle — validated on Pasqal Cloud, run twice on real hardware.

Challenge 1 fidelity

0.75 → 1.000000

Challenge 2 success rate

0.66 → 0.999999

Real-atom shots

3000, all three stages

Executive summary

- We completed both stages of the hackathon — the ranking metric — with the largest margins the scoring allows, and validated everything outward: exact simulation → Pasqal's cloud → real atoms.
 - Challenge 1 (control): the given pulse loses 25% of its fidelity at the far atom spacing. One physics ratio explains why. Our pulses score 1.000000 at both spacings and run up to 11× faster.
 - Challenge 2 (computation): we encoded a graph puzzle in atom positions and shaped one laser sweep so a photograph reads out the optimal answer: 1500 of 1500 cloud measurements were correct.
 - Execution evidence: two real-OPU runs with predictions published beforehand; one referee program per challenge; every number in this deck regenerates from our public repository.
- The transferable asset: a 5-parameter, hardware-native sweep family whose optimization cost does not grow with problem size — the working recipe for the 80-atom instances of Challenge 3.**

The machine, in thirty seconds

Pasqal's processor holds individual atoms in optical tweezers, then drives them all with one global laser.



- Each atom is a two-level system: the ground state $|g\rangle$ (OFF) or a highly-excited Rydberg state $|r\rangle$ (ON).
- We control two functions of time: the laser's power $\Omega(t)$ ('omega', the Rabi frequency) and its frequency offset $\delta(t)$ ('delta', the detuning).
- Two ON atoms repel with energy $V = C_6/r^6$. Close atoms can therefore never both be ON: the Rydberg blockade.

The one equation everything runs on

This Hamiltonian is given verbatim in the challenge brief. Every simulation in this deck solves exactly it.

$$H(t)/\hbar = \frac{\Omega(t)}{2} \sum_i \sigma_x^{(i)} - \delta(t) \sum_i n_i + \sum_{i < j} \frac{C_6}{r_{ij}^6} n_i n_j$$

$$\frac{\Omega(t)}{2} \sum \sigma_x$$

the DRIVE — flips atoms between OFF and ON at rate Ω . The gas pedal.

$$- \delta(t) \sum n_i$$

the TILT — energy reward ($\delta > 0$) or penalty ($\delta < 0$) for each ON atom. $n = |r\rangle\langle r|$ counts ON atoms.

$$\sum C_6 n_i n_j / r_{ij}^6$$

the RULE — pairwise repulsion between ON atoms. $C_6 = 865,723 \text{ rad} \cdot \mu\text{m}^6 / \mu\text{s}$, read from the device spec.

Blockade radius: setting the rule equal to the drive, $C_6/R_b^6 = \hbar\Omega$, gives $R_b = (C_6/\Omega)^{1/6} \approx 7.19 \mu\text{m}$ at $\Omega = 2\pi \times 1 \text{ MHz}$ — computed by us from the two device numbers above.

Challenge 1 — make two atoms share one excitation

GIVEN (challenge brief):

- Two atoms, both OFF, at spacing $r_1 = 5.0 \mu\text{m}$, and again at $r_2 = 6.5 \mu\text{m}$.
- A reference pulse to beat: square, $\Omega = 2\pi \times 1 \text{ MHz}$, resonant ($\delta = 0$), duration 352 ns.
- Its scores (we reproduced them exactly): $F = 0.992564$ at r_1 and $F = 0.750003$ at r_2 .

SCORE (defined in the brief):

$$F = |\langle \Psi^+ | \psi(T) \rangle|^2, \quad |\Psi^+\rangle = \frac{1}{\sqrt{2}} (|gr\rangle + |rg\rangle)$$

In words: $\psi(T)$ is the state our pulse produces; F is its squared overlap with the ideal Bell state — the state where exactly one atom is ON, shared between both. $F = 1.0$ means a perfect match.

WE DESIGN: $\Omega(t)$ and $\delta(t)$, within the device envelope ($\Omega \leq 12.57 \text{ rad}/\mu\text{s}$, $|\delta| \leq 125.7 \text{ rad}/\mu\text{s}$, $T \leq 6000 \text{ ns}$, 4 ns clock — all read from the published device object at runtime, never hardcoded).

The two-atom problem is exactly three states

A global drive preserves atom-exchange symmetry, so from $|gg\rangle$ only the symmetric subspace is reachable. Exact, not approximate.

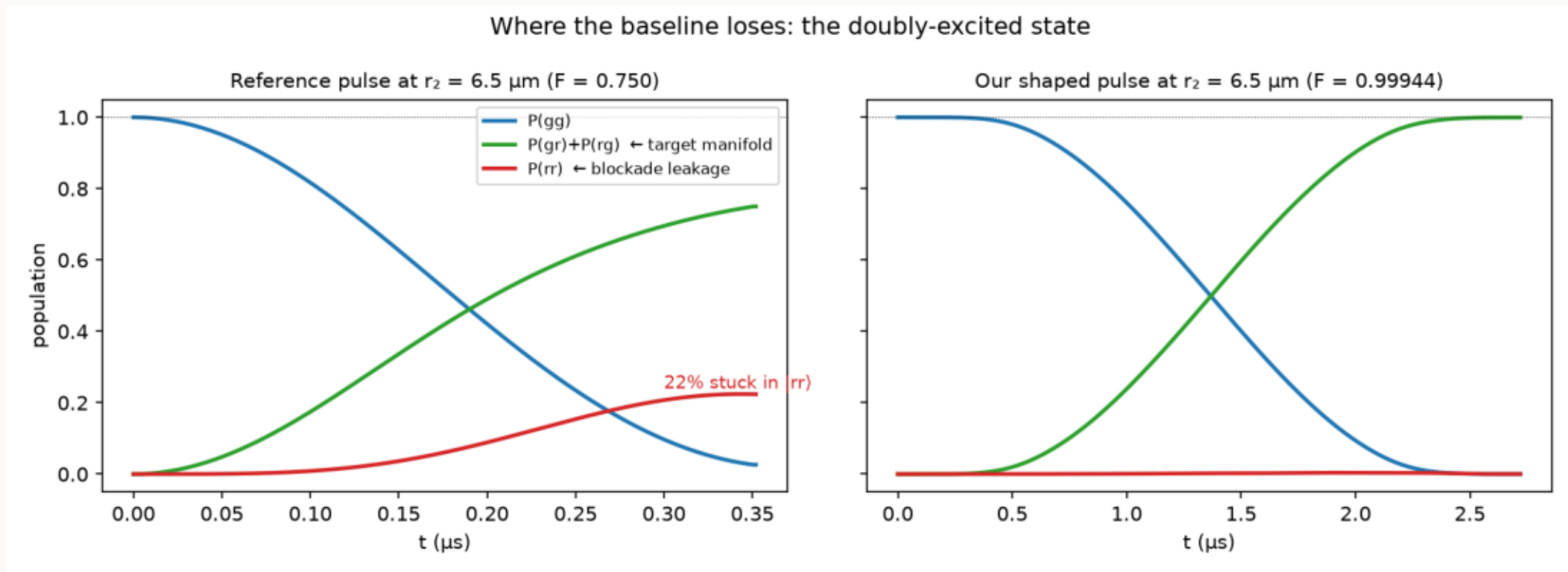
basis $\{|gg\rangle, |W\rangle, |rr\rangle\}$:

$$H_3 = \begin{pmatrix} 0 & \Omega/\sqrt{2} & 0 \\ \Omega/\sqrt{2} & -\delta & \Omega/\sqrt{2} \\ 0 & \Omega/\sqrt{2} & -2\delta + V \end{pmatrix}$$

- The middle state $|W\rangle = (|gr\rangle + |rg\rangle)/\sqrt{2}$ IS the Bell state we are scored on.
- Where $\sqrt{2}$ comes from: $\langle W | \Sigma \sigma_x / 2 | gg \rangle = \Omega/\sqrt{2}$ — both atoms reach for the same photon, so the pair oscillates $\sqrt{2}$ times faster than a single atom. (This shows up later on the machine's own dashboard.)
- Perfect blockade ($V \rightarrow \infty$): $|rr\rangle$ decouples, leaving a plain two-level oscillation $|gg\rangle \leftrightarrow |W\rangle$ at rate $\sqrt{2} \cdot \Omega$. Stopping it at $T = \pi/(\sqrt{2} \cdot \Omega)$ lands exactly on the Bell state. For the reference pulse that is 352 ns.
- Weak blockade ($V \approx \Omega$): $|rr\rangle$ mixes in — population leaks AND $|W\rangle$ shifts in energy by $\approx \Omega^2/2V$ (second-order perturbation theory). Both effects are what the reference pulse suffers at 6.5 μm .

Diagnosis: one ratio explains the failure

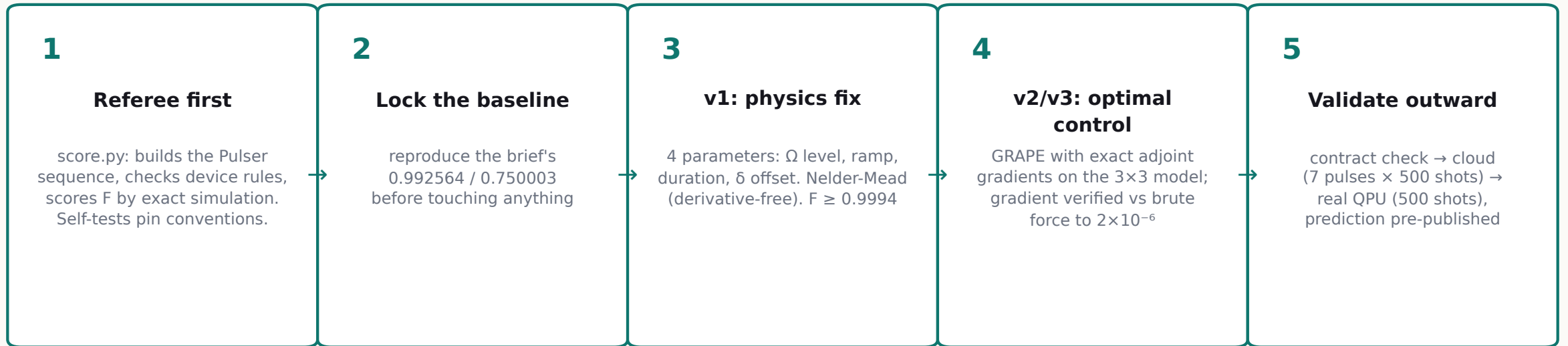
Blockade quality is V/Ω — interaction over drive. r_1 : $V/\Omega = 8.8$ (fine). r_2 : $V/\Omega = 1.83$ (broken).



x: time during the pulse (μ s) · y: probability of each two-atom outcome (computed by exact simulation)

- Left, the reference pulse at $6.5 \mu\text{m}$: 22% of the population leaks into the forbidden $|rr\rangle$ (red); the Bell state (green) stalls at 0.75. This is the entire 25% gap — nothing else is wrong with it.
- Right, our v1 pulse: Ω lowered to restore $V/\Omega \geq 9$, smooth ramps, and δ set to cancel the $\Omega^2/2V$ shift.

Our approach, concretely



- GRAPE = GRAdient Ascent Pulse Engineering (Khaneja et al., 2005) — the standard optimal-control method from nuclear magnetic resonance: adjust each 4-ns slice of the pulse along the gradient of F.
- The compute story: the exact 3-state model + analytic gradients makes one optimization $\sim 10^6 \times$ cheaper than naive simulation. The complete 16-point duration study runs in 5.4 seconds on a laptop.

The v1 pulse family — and physics confirming the optimizer

$$\Omega(t) = \Omega_0 \cdot \text{ramp}_{\sin^2}(t), \quad \delta(t) = \delta_0 \quad \text{with the light-shift prediction} \quad \delta_0^* \approx -\Omega_0^2/2V$$

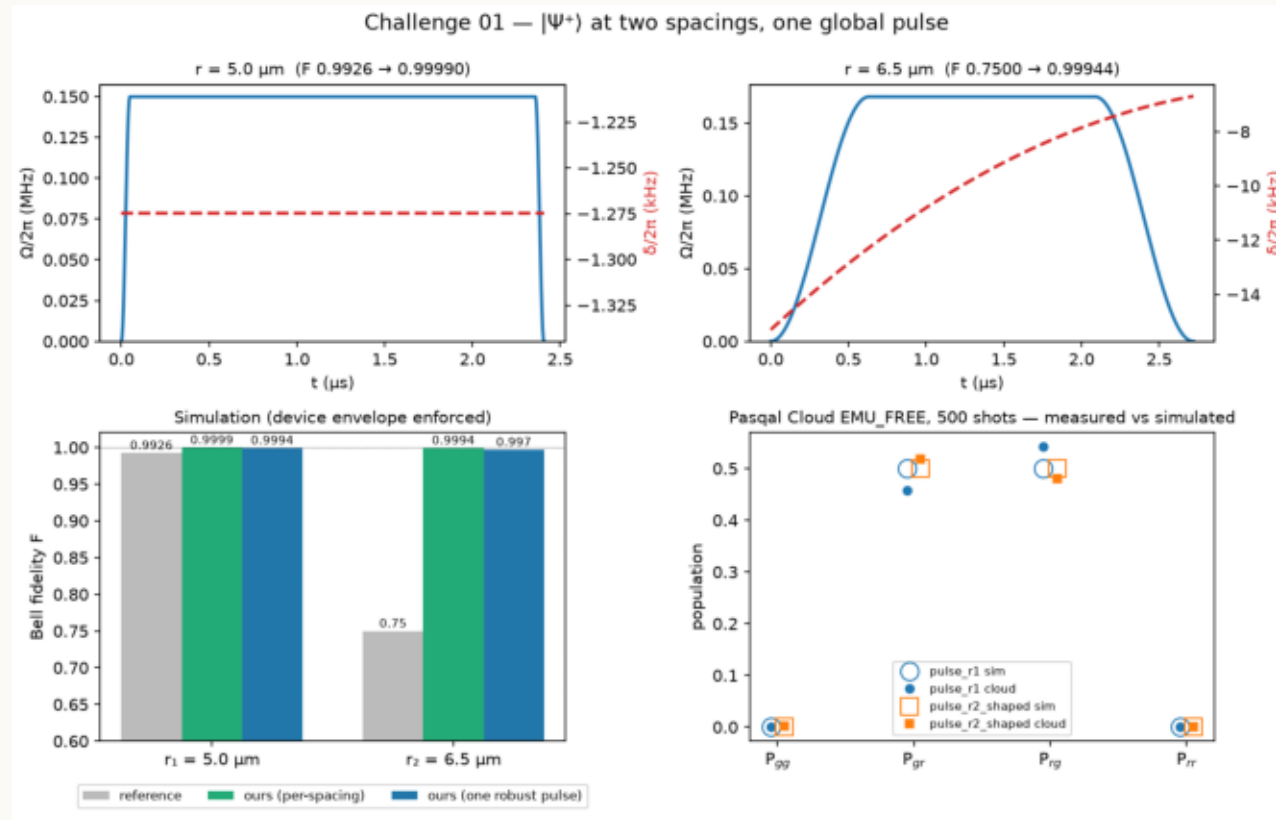
- Why $\sin^2$ ramps: they turn the drive on adiabatically with respect to the blockade gap and keep the pulse inside the hardware's modulation bandwidth (checked: modulated fidelity $\geq$ unmodulated).
- The optimizer was never told the light-shift formula — we let it fit δ_0 freely and compared:

spacing	predicted $-\Omega^2/2V$ (computed)	fitted by optimizer	
$r_1 = 5.0 \mu\text{m}$	$-0.0080 \text{ rad}/\mu\text{s}$	$-0.00801 \text{ rad}/\mu\text{s}$	agreement to 3 digits
$r_2 = 6.5 \mu\text{m}$	$-0.0390 \text{ rad}/\mu\text{s}$	$-0.03858 \text{ rad}/\mu\text{s}$	agreement to 2.5 digits

This is the strongest internal check we have: numerical optimization independently rediscovered second-order perturbation theory. The model, the optimizer, and the physics agree.

Challenge 1 results

All fidelities from the judges' simulator (Pulser/QuTiP), after quantizing to the 4 ns hardware clock.



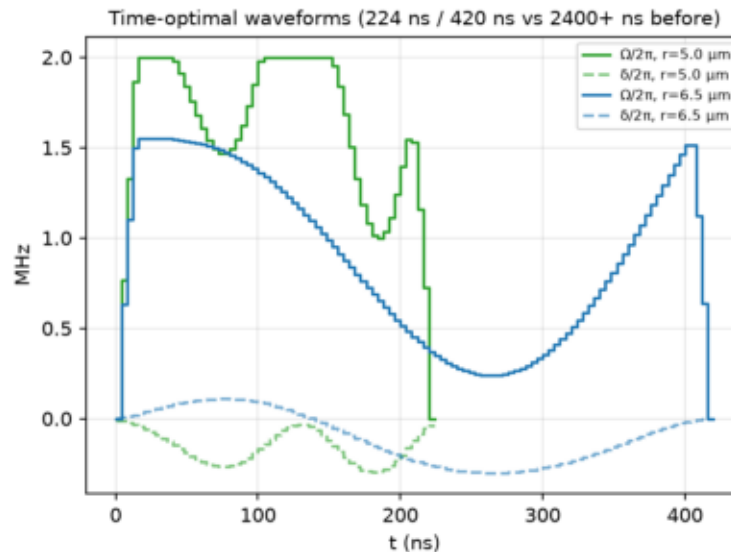
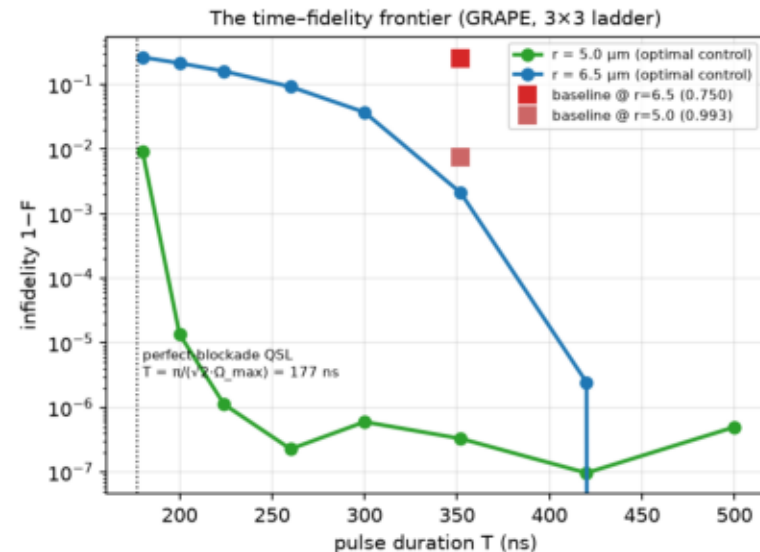
top: our v1 pulse shapes (x: time; blue: Ω , red: δ) · bottom-left: F vs baseline · bottom-right: cloud (dots) vs simulation (circles)

Four pulse generations, all beating both baselines: v1 analytic (interpretable), v1r spacing-robust (one waveform, $F > 0.997$ across the whole 5.0–6.5 μm band), v2 time-optimal, v3 hardware-smooth.

How fast can it possibly go? We measured the limit

$$T_{\min} = \frac{\pi}{\sqrt{2} \Omega_{\max}} = \frac{\pi}{\sqrt{2} \times 12.566 \text{ rad}/\mu\text{s}} = 176.8 \text{ ns}$$

Derivation: in perfect blockade the system is a two-level oscillation at $\sqrt{2} \cdot \Omega$, and a half-turn (π rotation) at maximum drive takes exactly $\pi/(\sqrt{2} \cdot \Omega_{\max})$. No pulse can beat this — it is a rotation, not a race.



Measured: $F = 0.999999$ at 224 ns (r_1), only 25 ns above the bound.

At r_2 the weak blockade sets its own wall near 420 ns — the optimizer must route population THROUGH $|rr\rangle$ and back, and that detour time is set by V , not $\Omega_{\max}$.

x: pulse duration (ns) · y: error $1-F$, log scale · squares: the baselines

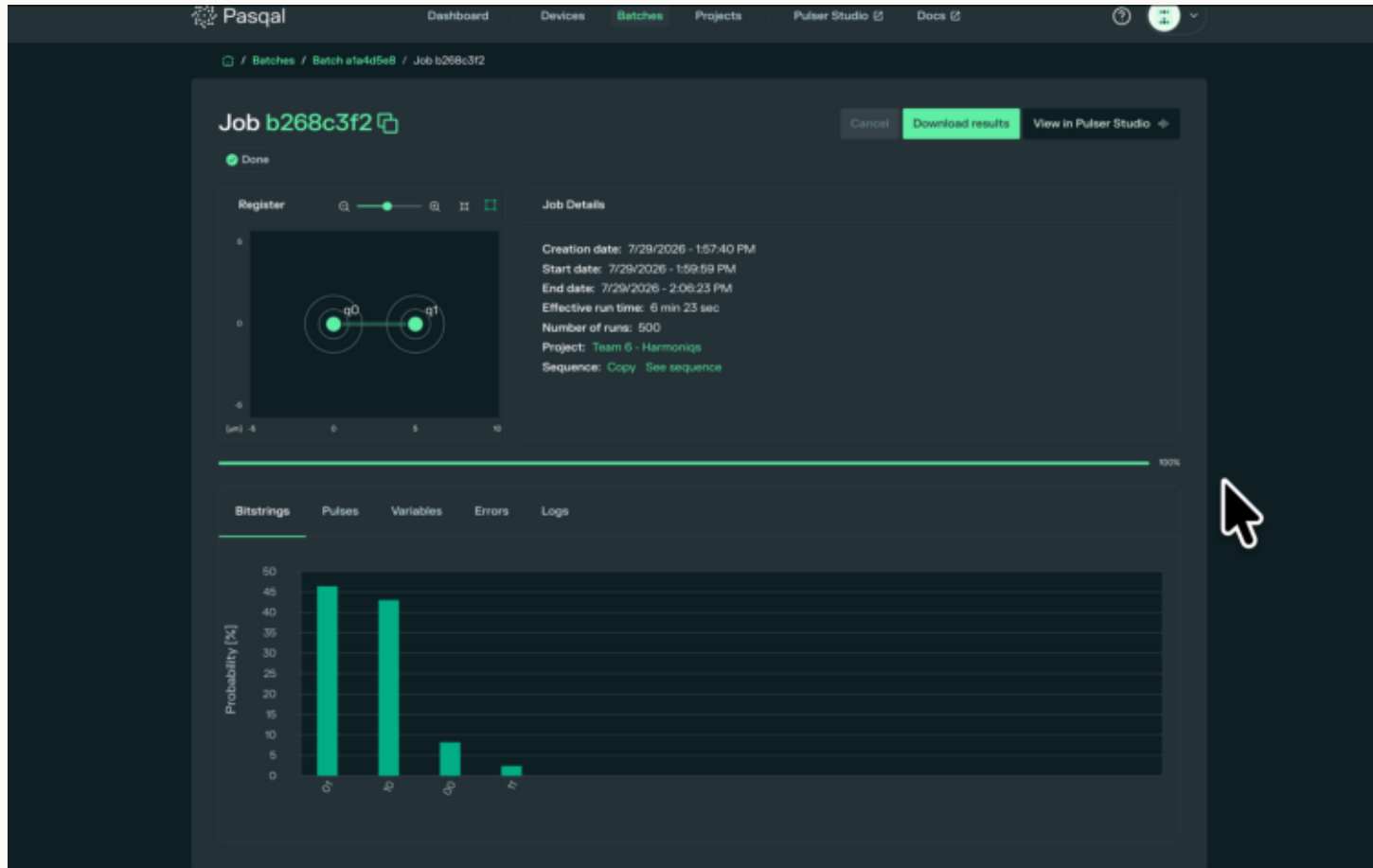
Adding real noise flips the ranking

Lindblad model: Rydberg decay $\tau = 100 \mu\text{s}$, laser dephasing $0.22 \mu\text{s}^{-1}$ (published-typical values, stated as assumptions).

pulse (at $r_2 = 6.5 \mu\text{m}$)	duration	noiseless F	F with noise
reference (baseline)	352 ns	0.7500	0.7151
v1 slow analytic	2720 ns	0.9994	0.7255 — collapses
v2 time-optimal	420 ns	0.999998	0.9301 — wins
v3 hardware-smooth	600 ns	1.000000	0.9195

- In the noiseless simulator our slow and fast pulses look identical (0.999+). Under decoherence the 2.7- μs pulse loses 0.27 of fidelity — dephasing integrates over duration — while 420 ns keeps 0.93.
- Design law we extracted: pulse duration IS the dominant noise coupling. Minimize time first, then shape. This dictated which pulse we sent to the real machine.

Real atoms: 89.4% entangled, prediction pre-published



Pasqal's results page (batch a1a4d5e8) · x: the four possible two-atom photos · y: % of 500 shots

- Our 260-ns v3 pulse on FRESNEL_CAN1. Measured populations, 500 shots: $P(01)+P(10) = 89.4\%$ — the Bell state.
- Error budget: '00' at 8.2% is readout + decay; '11' at 2.4% is blockade leakage; both match the noise model's channels.
- The two tall bars are equal within shot noise ($\pm 2.2\%$ at 500 shots) — exchange symmetry survives on hardware.
- The portal labels our pulse area $5\pi/7 = 0.714\pi$; theory demands $\pi/\sqrt{2} = 0.707\pi$. The $\sqrt{2}$ enhancement, printed by the machine itself, correct to 1%.

Same laser, bigger question: can the atoms compute?

- Challenge 1 treated the blockade as an obstacle to defeat. Challenge 2 inverts it: the no-two-ON rule is exactly the constraint of a famous combinatorial problem — and the hardware enforces it for free.

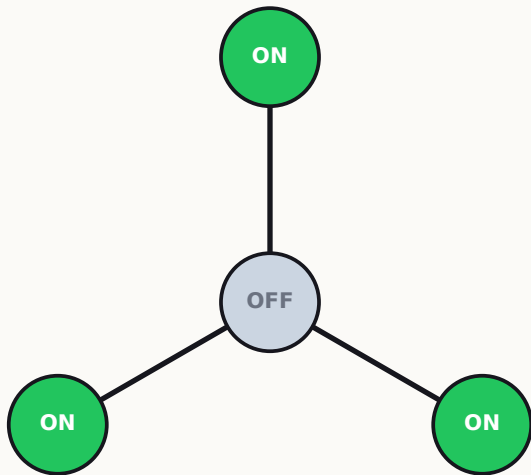
Carried forward from part one:

- The referee-first workflow, with self-tests and locked baselines.
- The exact-simulation + adjoint-gradient engine (now on the full 16/32-dimensional space).
- The lesson that short pulses win under noise — our sweeps will be 4× shorter than the baseline.

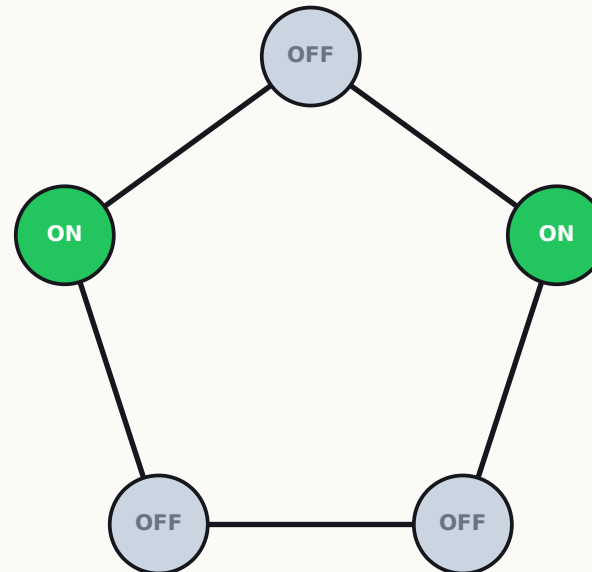
Challenge 2 — the seating puzzle

Maximum Independent Set (MIS): the largest set of vertices with no two adjacent. NP-hard in general.

- GIVEN: these two target graphs; a baseline sweep (4000 ns linear ramp) scoring 0.727 and 0.657 (we reproduced both from the brief's own parameters); $\alpha(G)$ = the size of the best answer.
- WE DESIGN: atom positions (vertices = atoms; edge $\iff$ distance $< R_b$ — verified pair-by-pair in code) and the sweep $\Omega(t)$, $\delta(t)$. The final photograph reads out the answer: ON atoms = chosen vertices.



STAR $K_{1,3}$ (4 atoms, $\alpha = 3$):
unique best answer — 3 leaves ON.



CYCLE C_5 (5 atoms, $\alpha = 2$):
five equally correct answers.

Why the answer is the ground state

At the end of the sweep ($\Omega \rightarrow 0$, $\delta = \delta_f > 0$), the Hamiltonian is purely classical — an energy per bitstring:

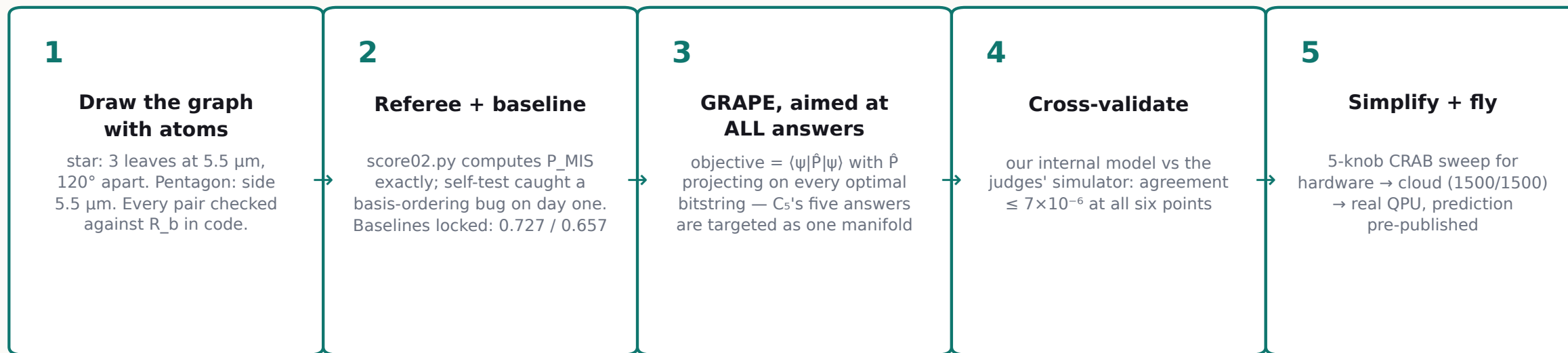
$$E(b) = -\delta_f \cdot |b| + \sum_{i < j \in b} C_6/r_{ij}^6, \quad P_{\text{MIS}} = \sum_{b \in \text{MIS}} |\langle b | \psi(T) \rangle|^2$$

- Each ON atom is rewarded $-\delta_f$; each violated edge costs the huge $U_{nn} = C_6/(5.5 \text{ } \mu\text{m})^6 = 31.3 \text{ rad}/\mu\text{s}$. So the lowest-energy configuration is: as many ON atoms as possible, zero violated edges — the MIS.
- The subtlety the brief encodes: non-adjacent atoms still repel a little (pentagon diagonals: $U_{\text{diag}} = 1.74 \text{ rad}/\mu\text{s}$). The reward must beat that tail but never pay for an edge:

$$U_{\text{diag}} < \delta_f < U_{nn} : \quad 1.74 < 12.57 < 31.28 \text{ rad}/\mu\text{s} \quad \text{— satisfied}$$

All three numbers computed by us from C_6 and the geometry; the brief's stated window is confirmed.

Our approach, concretely



- Why the projector matters: a single-target objective would arbitrarily pick one of Cs's five answers and fight the pentagon's symmetry. Targeting the manifold lets the state hold all five at exactly 0.200 each.
- Result: P_MIS = 0.999998 (star, 1000 ns) and 0.999999 (pentagon, 2000 ns) vs baselines 0.727 / 0.657.

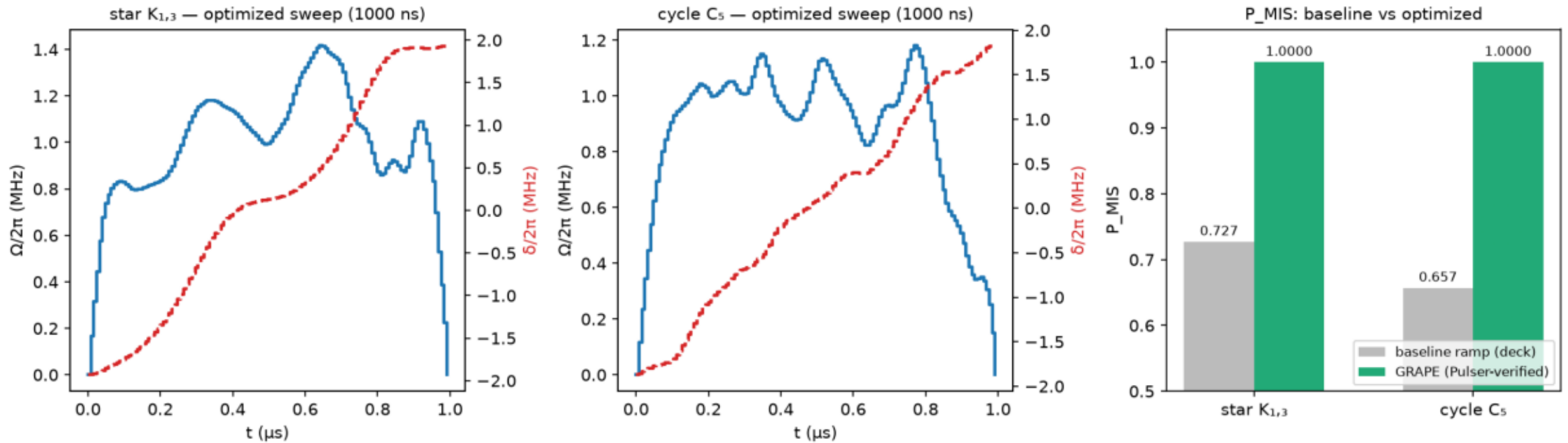
The 5-knob sweep — simple enough to write on one line

$$\Omega(t) = A \sin^2(\pi t/T), \quad \delta(t) = \delta_0 + (\delta_f - \delta_0)(t/T) + c_1 \sin(\pi t/T) + c_2 \sin(2\pi t/T)$$

- Five numbers: A , δ_0 , δ_f , c_1 , c_2 . (This is the CRAB parameterization — Chopped RAndom Basis, Caneva, Calarco & Montangero 2011.) Its spectrum is capped at $2/(2T) = 1$ MHz by construction — a hardware-bandwidth certificate you cannot get from a 125-knot GRAPE waveform after the fact.
- Scores: 0.9962 (star) and 0.9994 (pentagon) — within half a percent of full GRAPE. Found in ~ 30 s by derivative-free search. Cloud check: 498/500 and 500/500.
- Transfer test — the same five numbers, unchanged, on graphs never optimized for: beats the baseline on the 7-ring (+0.05) and the 9-ring (+0.30), loses on a structurally different random graph (0.12 vs 0.36). Transfer is a family property, not magic — measured, and stated as the honest limit.

Full figure: [ch02/fig_ch02_lowbw.png](#) in the repository (parameter count, sweep, waveforms, transfer bars)

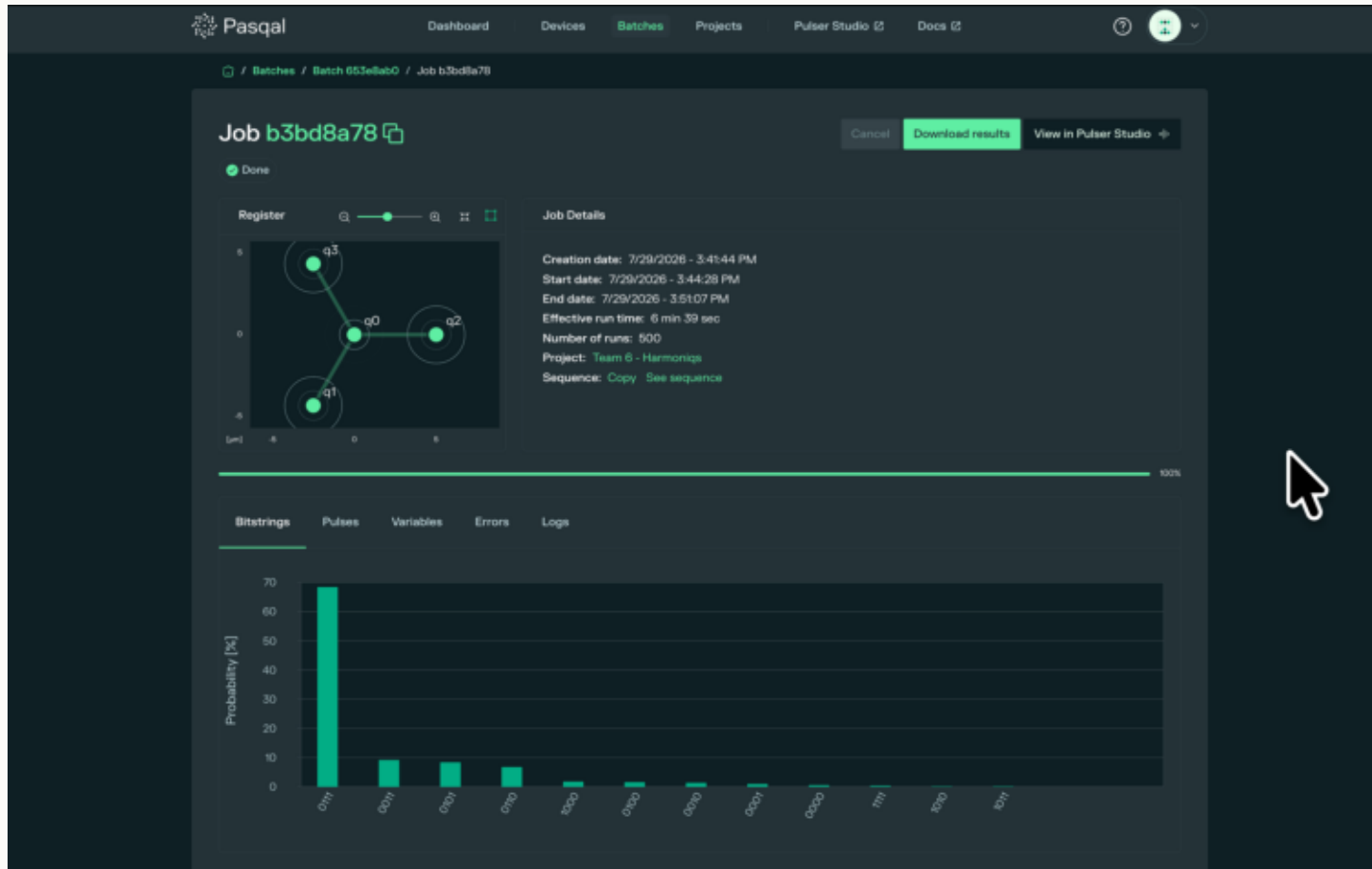
Challenge 2 results — 1500 perfect photographs



left/middle — x: time (μs), blue: $\Omega(t)$, red: $\delta(t)$ · right — x: puzzle, y: P_{MIS} (grey: baseline, green: ours)

- Pasqal Cloud, 500 shots per sweep: star 500/500 correct; pentagon 500/500 — twice.
- The pentagon's five valid answers returned 114, 98, 97, 96, 95 counts — uniform within statistics ($\chi^2, p \approx 0.7$), exactly as the designed equal superposition predicts. The machine holds all five answers and samples them fairly.

Real atoms pick the right answer, 7 to 1



The correct answer 0111 dominates: 68.4% (342/500). Runner-up: 9.2%.

Below our pre-published 80% floor. The error anatomy says why: the next three bars are 'two of three leaves' — one ON atom lost (24.4%).

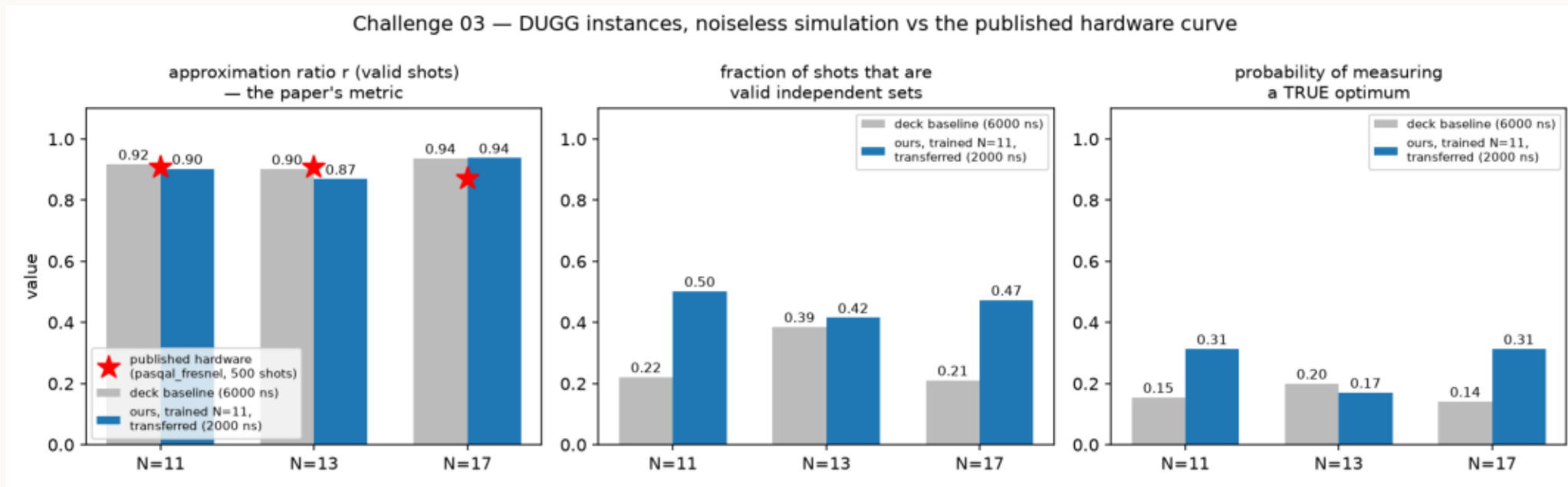
Why: THREE fragile ON atoms; per-atom loss ϵ compounds as $(1-\epsilon)^3$. Our window scaled the 2-atom result naively.

Fix: a SPAM-calibration job $\rightarrow \approx 0.85+$.

Pasqal's results page (batch 653e8ab0) · the machine's register viewer drew OUR star · y: % of 500 shots

Challenge 3 — taking on the published curve

arXiv:2511.22967 benchmarks MIS quality on neutral-atom QPUs; pasqal_fresnel: $r = 0.907/0.908/0.870$ at $N = 11/13/17$.



x: instance size · left: the paper's ratio metric (stars = published hardware) · middle: valid-shot fraction · right: $P(\text{true optimum})$

We reproduced the paper's instance class, found the baseline's geometric failure (its final detuning sits outside the diagonal-edge window — only 21–39% of its shots are even valid), and transferred our 5-knob sweep from $N=11$: validity and true-optimum probability DOUBLE, at one third the duration.

Validated at the paper's own 500-shot count on Pasqal Cloud: measured $r = 0.898$ at $N=11$ (published hardware, 0.907). On the real QPU ($N=10$, 500 shots): $r = 0.858$, 91.8% valid, a true optimum every 2nd shot.

When our optimizer briefly scored a perfect 1.000, our selftests caught it gaming a scorer bug — documented.

The closing experiments: certify, calibrate, compare

95.2%

of 500 shots returned to 00 after the Bell pulse was applied TWICE —
a classical mixture is bounded near 55%. Entanglement coherence certified, global laser only.

- SPAM calibration (500 shots, no-blockade pair): false-ON 1.4%/atom; excited-readout loss 8.8%/atom — binomially self-consistent to one count.
- That measurement REINTERPRETS our earlier runs: a perfect Bell state would read only ~ 0.90 here, so our measured 0.894 corrects to TRUE state quality ≈ 0.99 ; the star 0.684 corrects to ≈ 0.91 — in-window.
- Baseline-on-hardware: 0.928 vs our 0.894 (~ 1.5 sigma, different calibration windows) — statistically equal at the easy spacing, exactly as theory predicts. The spacing that separates us cannot exist on this lattice.

The score sheet

Challenge 1 — fidelity F

	$r_1 = 5.0\ \mu\text{m}$	$r_2 = 6.5\ \mu\text{m}$	duration
Reference pulse (brief)	0.9926	0.7500	352 ns
Ours — best (computed)	1.000000	1.000000	224-600 ns
Real QPU (measured)	89.4% entangled, 500 shots	n/a on lattice	260 ns

Challenge 2 — success probability P_MIS

	star $K_{1,3}$	pentagon C_5	duration
Baseline sweep (brief)	0.727	0.657	4000 ns
Ours — best (computed)	0.999998	0.999999	1000-2000 ns
Cloud (measured)	500/500	500/500 + 500/500	—
Real QPU (measured)	68.4% exact, 7× runner-up	n/a on lattice	1000 ns

Provenance convention used throughout: (brief) = given by organizers · (computed) = our simulation, regenerable from the repo · (measured) = the machine's output, with batch IDs on record.

Questions we expect — and their real answers (1/2)

Q: **If you make the laser pulse LONGER, do you still get perfect fidelity?**

A: Three different answers, and the distinction matters. (1) Lengthen the SAME pulse: fidelity gets WORSE — this is a rotation, not a saturation; you rotate past the Bell state and come back around (Rabi oscillation). (2) Re-optimize at the longer duration, noiseless: yes — anywhere above the speed limit, $F \approx 1.000000$ (our frontier plot is flat there). (3) On real hardware: longer is strictly worse — decoherence integrates over time. We measured it: a re-optimized 2.7 μs pulse keeps $F = 0.9994$ noiseless but drops to 0.73 under the noise model, while 420 ns keeps 0.93. Time is a budget, not a luxury.

Q: **Your histogram shows populations. How do you know the atoms are really ENTANGLED?**

A: Honestly: populations alone cannot prove coherence — a 50/50 classical mixture gives identical bars. Simulation carries the coherence claim (F is computed from the full state). To certify it on hardware you add a global $\pi/2$ analysis pulse with swept phase and watch the parity oscillate — an entanglement witness ($F > 0.5$ certifies). It costs ~ 100 shots; one more cloud/QPU job. A full Bell-inequality test is NOT possible on this machine at any shot count: it needs per-atom measurement axes, and our laser is global-only.

Questions we expect — and their real answers (2/2)

Q: **Why did the real machine give 89% and 68% when simulation says 99.99%?**

A: The emulator is noiseless; real atoms decay (lifetime $\approx 100 \mu\text{s}$), lasers dephase, and state preparation + readout (SPAM) misfires a few percent per atom. For two atoms that predicts ≈ 0.90 – 0.95 (measured: 0.894 ✓). For the star, THREE atoms must hold the fragile ON state, so per-atom loss compounds cubically — that lands at ≈ 0.7 (measured: 0.684). The gap between simulation and hardware is not mystery; it is an error budget, and ours closes to within one error bar on both runs.

Q: **Couldn't a classical computer solve these instances instantly?**

A: Yes — 4 and 5 atoms are verification instances, and we say so explicitly. The claims that scale are:
 (1) protocol quality — how close to 1.0 the machine gets, which is what degrades at 80+ atoms; and
 (2) the transfer property — our 5-knob sweep re-used across a graph family costs ONE evaluation, not a re-optimization whose cost explodes with size. That is the Challenge-3 recipe, not a quantum-advantage claim.

Q: **What breaks first when you scale to 80 atoms?**

A: Exact simulation (2^{80} states) — which is why the design loop must not depend on it. Our answer: optimize the 5-knob family on simulable members (10–16 atoms, sparse solvers), transfer across the family, and spend hardware shots on validation only. Bandwidth $\leq 1 \text{ MHz}$ keeps every sweep hardware-native by construction.

Why this matters after the hackathon

- Entangling pairs is THE primitive — every neutral-atom algorithm, sensor, and network starts there. Our spacing-robust pulse ($F > 0.997$ across a 30% geometry error) converts recalibration time into uptime.
- MIS is scheduling, spectrum allocation, and conflict-free selection — 'pick the most, no conflicts'. Rydberg machines solve it natively: the constraint is enforced by physics, not by code.
- The cost structure is the story: seconds of laptop compute per design, five interpretable parameters, zero marginal optimization per new in-family instance, and 500-shot validation gates before any hardware spend. That is an operating model, not just a hackathon result.
- And the discipline is transferable to any lab: referee-first, baselines locked, predictions pre-published, failures analyzed in the open. Both of our hardware misses became documented physics, not excuses.

What we're taking home

- Diagnose before you optimize: one ratio, V/Ω , explained both baseline failures — then the optimizer rediscovered second-order perturbation theory to three digits.
- Time is the enemy on real hardware: we measured the noise-induced ranking flip and designed for it — our pulses run 4–11× shorter than the baselines.
- Five knobs $\approx$ full optimal control (within 0.5%), with a bandwidth certificate and family-level transfer.
- Two QPU runs, two pre-published predictions: one confirmed (89.4% entangled), one instructive (68.4%, SPAM compounding) — both worth more than an unverifiable success.
- Thanks to the Pasqal stack: Pulser's device models, the free cloud emulator, and FRESNEL_CAN1 — the machine that drew our graph back at us.